Sex Differences in Psilocybin Acute and Sub-Acute Therapeutic Effects on Depressive and Anxious Symptoms in Rats

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This experiment was supervised by Bart Ellenbroek, and all data was collected by Henry Chafee and Larissa Rocha from the Victoria University of Wellington Behavioural Neurogenetics Lab.

Abstract

Background Clinical studies support psilocybin as a viable option for the treatment of affective disorders. There are few preclinical findings on this topic, and those available have been inconclusive, and exclude females. No one to our knowledge has reported on sex differences in therapeutic effects of psilocybin in rats, though an inhibited acute response by females has been reported. The current experiment adds to this research by testing for a therapeutic effect of psilocybin on anxious and depressive symptoms in male and female rats, as well as any sex effects in the acute response and sub-acute therapeutic outcomes.

Method Early maternal separation was used to induce anxious and depressive symptoms, which were tested by the novel Affective Disorders Test (ADT). Subjects were treated with a single intraperitoneal injection of either saline, 8 or 16 mg/kg of psilocybin. Three further days of ADTs were conducted post-treatment. Based on past findings, we predicted a therapeutic effect of psilocybin, an inhibited acute response to psilocybin in female subjects, and an inhibited therapeutic effect in female subjects.

Results No significant sex differences in acute responses to psilocybin were detected. We were able to detect a therapeutic effect of psilocybin within our sample, though this effect was strongly driven by the females' much stronger therapeutic response.

Conclusion We concluded that psilocybin has a therapeutic effect on anxious and depressive symptoms in rats, offering support to clinical studies. We also concluded that female rats are more sensitive to this therapeutic effect than males, though several possible reasons for this are given, including a floor effect in our male due to a weaker response to EMS. We could not disconfirm the presence of a sex effect in acute psilocybin response, due to low statistical power and additional limitations of this experiment. We instead suggest that separation of females by oestral phase, and behavioural tests during treatment may be necessary to detect this effect.

Introduction

Affective Disorders

Depression and anxiety, collectively referred to as affective disorders, are extremely prevalent psychological disorders. They commonly co-occur and can have extreme outcomes such as complete incapacity and death by suicide (American Psychological Association, 2013). Major Depressive Disorder is the most common form of depression described in the Diagnostic and Statistical Manual Fifth edition (DSM-5), which involves multiple major depressive episodes (MDEs) throughout the lifetime. The DSM-5 diagnostic criteria for MDEs include clear affective, cognitive, and physical changes such as hopelessness, fatigue, difficulty concentrating, sleep and weight disturbances, decreased ability to experience pleasure (anhedonia), and suicidal thoughts (American Psychological Association, 2013). The average world-wide prevalence rate of MDD in 2017 was 3.4% (Ritchie & Roser, 2018), however lifetime prevalence rates reach as high as 21% in some countries (Kessler & Bromet, 2013; Gutiérrez-Rojas et al., 2020).

Generalized Anxiety Disorder (GAD) is the most common form of anxiety disorder described in the DSM-5 (American Psychological Association, 2013). Patients experiencing this disorder may suffer from excessive worry, restlessness, sleep disturbance, difficulty concentrating or mind going blank, muscle tension and fatigue (American Psychological Association, 2013). An estimated 33.7% of the population worldwide experiences an anxiety disorder at least once in their lifetime (Bandelow & Michaelis, 2015), and anxiety disorders can cause moderate to severe disability, accounting for an estimated 110 million missed workdays each year in the United States (American Psychological Association, 2013). Additionally, the comorbidity rate for depressive and anxious symptoms is estimated at 85-90% (Gorman, 1996), with comorbidity estimates specific to GAD and MDD reaching as high as 72% (Zhou et al., 2017). Additionally, every dollar invested in the treatment of depression and anxiety is estimated to result in a $4 return to the economy, making this not only an important issue for those suffering from these disorders, but also for the global economy (American Psychiatric Association, 2014).

Current Treatments

Currently, the most effective treatments of affective disorders are cognitive behavioural therapy (CBT) and antidepressant medications, the most common type being selective serotonin reuptake inhibitors (SSRIs) (Rush et al., 2009; Baldwin et al., 2011). CBT focuses on retraining the brain to favour more positive thought patterns and emotional responses (Hazlett-Stevens & Craske, 2002), while SSRIs increase serotonin (5-HT) in the brain (Cowen & Browning, 2015; Nutt et al., 1999). Increased serotonin levels impact anxiety and depression by promoting neural plasticity and hippocampal neurogenesis, improving learning and memory function, while also encouraging adaptations in behaviour (Cowen & Browning, 2015; Alenina & Klempin, 2015; Branchi et al., 2013). The combination of psychotherapeutic and pharmacotherapeutic treatments has consistently shown the largest positive impact on long-term symptomatology (Cuijpers et al., 2020; Zobel et al., 2011; Cuijpers et al., 2009). However, due to the time-intensive nature of psychotherapy, high cost of private psychotherapy, and demand exceeding the supply of trained psychologists, those who pursue psychological treatment are required to wait weeks, even months before receiving treatment (Walton, 2003; New Zealand Government, 2018). As a result, antidepressant medications alone have become the most common treatment of depression (National Institute for Health and Care Excellence, 2015; Malhi et al., 2015). Demonstrating their common use, nearly 9% of New Zealanders were prescribed antidepressants in 2020 (Ministry of Health, 2021).

Despite this common use of antidepressants, however, rates of response and remission resulting from antidepressant medications such as SSRIs are far from ideal, and the rates of drop-out seen are significantly higher than in psychotherapy or combined treatment (Cuijpers et al., 2020). The Sequenced Treatment Alternatives to Relieve Depression (STAR*D) conducted in 2009, which is the largest study on the efficacy of antidepressant medications to date (n= 4041), illustrated these issues. A cumulative remission rate of only 67% across four 12-week treatments was found in this study, with only 28% achieving remission during the first treatment trial (Rush et al., 2009). While these numbers are already low, Pigott (2015) made the critique that these rates do not account for drop-out rates, and that when they are accounted for, this remission rate drops only 45.9% (Pigott, 2015). In fact, more participants dropped out than remitted in each treatment stage of the STAR*D study. This is presumably due to the adverse side-effects common to SSRIs, which have a significantly faster onset than the therapeutic effects. These side-effects include dizziness, weight gain, nausea, vomiting, headaches, insomnia, and sexual dysfunction (Kupfer, 2005; Demyttenaere & Jaspers, 2008). While these side-effects are experienced immediately, there is a 4–6-week delay before the onset of therapeutic effects (Cowen & Sargent, 1997). This early period in treatment can be dangerous, as patients may experience increased motivation without any improvements in mood, leading to a spike in suicide risk (Simon et al., 2006). Following this period, an estimated 20% of patients do not respond to SSRI treatment, despite up to two years of differing medications and doses (Kupfer, 2005). Those who do not respond following at least three months of SSRI treatment fall into the category of patients with treatment-resistant depression (TRD) (Rush et al., 2009).

Alongside SSRIs, Benzodiazepines (BZs) such as lorazepam and diazepam are commonly used to treat anxiety disorders, with prevalence rates of long-term BZ use estimated between 2.2 and 17% in the general population (Vicens et al., 2011). These too come with side-effects such as dependence, withdrawal, and psychomotor and cognitive impairment due to sedation (Lader, 2008). Accounting for BZ treatment dropout rates, which reach as high as 50% due to adverse side-effects, meta-analyses have at times found no evidence that BZs treat anxiety disorders more effectively than placebo (Martin et al., 2007; Shinfuku et al., 2019). A meta-analysis of 27 studies in 2011 reported that fluoxetine (an SSRI) was the most effective pharmacological treatment for anxiety. Despite this, response and remission rates were only 62.9% and 60.6% respectively (Baldwin et al., 2011). It was also unclear whether this figure accounted for the high dropout rates.

Therapeutic Potential of Psychedelic Substances

Further research is evidently required to establish alternative pharmacological treatments that can overcome these issues or be used alongside the current treatments of anxiety and depression for increased efficacy. Research on psychedelic substances have shown promising results in treating a range of psychological disorders including depression, TRD, anxiety, substance use disorders (SUDs) and PTSD.

Brief History of Psychedelics

Psychoactive substances such as psilocybe mushrooms, ayahuasca, mescaline and dimethyltryptamine (DMT)-derived substances have long histories and significant spiritual meaning within shamanistic cultures all over the world. Amazonian shamanistic cultures view psychoactive plants as a tool to "harmonize the individual with the surrounding world," (Bouso & Sánchez-Avilés, 2020, p.146) and have used them in coming-of-age ceremonies, spiritual rituals, sacrament, and to strengthen community bonds (Bouso & Sánchez-Avilés, 2020). Archaeological evidence of psychoactive alkaloids in artefacts and skeletal remains, as well as artistic depictions inspired by psychedelic experiences suggest their ritualistic use by many cultures since prehistoric times (Guerra-Doce, 2015; Samorini, 2001; Tupper et al., 2015; Wasson, 1961; Winkelman & Winkelman, 2019). Some representations of shamanic psychedelic rituals date back thousands of years, with the oldest representations suggesting the spiritual importance of psilocybe mushrooms in the form of African petroglyphs dating back to 7,000-9,000 B.C. (Guerra-Doce et al., 2015; Samorini, 2001). The use of these substances has been notably suppressed over the last 300 years, with the spread of Christianity and the inquisition of the New World, though some shamans have continued to practise these ancient rituals (Shepard, 2005). Classic psychedelics attracted a substantial amount of attention from the scientific community in the mid-20th century, however, following Albert Hofmann's synthesis of lysergic acid diethylamide (LSD) in 1938 (Hofmann, 1980). Studies run by Sandoz lab, who Hofmann conducted research on behalf of, described the possible psychotherapeutic uses of LSD, including its ability to "elicit [the] release of repressed material and provide mental relaxation, particularly in anxiety states and obsessional neuroses" (Hofmann, 1980, p. 29). Later in his career, Hofmann successfully isolated the active component in psilocybe mushrooms, psilocybin, which belongs to the indole family of natural substances, as does LSD (Hofmann, 1980).

Clinical Studies

Following a three-decade lull of publications on psychedelics due to restrictions on their production, this area of study has seen a recent resurgence, producing promising evidence for their efficacy in treating a wide range of psychological disorders. There is still, however, much ground to cover. A meta-analysis (n = 117) on the effects of psilocybin-guided therapy on anxious and depressive symptoms showed large decreases in symptoms three to four weeks following a single treatment session, which were sustained at the 6-month follow-up (Goldberg et al., 2020). However, the sample sizes included in this analysis were small. There was also a significant risk of outcome bias, due to limited blinding of participants to the hypotheses, and the studies included were heterogeneous in methodological design. A 2020 meta-analysis of nine randomized, placebo-controlled clinical trials supported this finding, however. In these studies, patients received one to three sessions of psychedelic-assisted therapy to assess its effects on PTSD, MDD, anxiety/depression caused by terminal illness, and comorbid social anxiety and autism (n = 211). This analysis showed a very large difference between treatment and placebo groups' outcomes (Hedges *g* = 1.21), representing an 80% probability that a randomly selected participant from the treatment condition of these studies had a better outcome than a randomly selected participant from the placebo condition (Luoma et al., 2020). These studies were more homogenous in design, though they still suffer from potential bias due to lack of blinding, and small sample sizes. One placebo-controlled crossover trial including 29 anxious and/or depressed terminally ill patients found immediate, clinically significant reductions in depressive and anxious symptoms, sustained by 60-80% of participants six and a half months following a single psilocybin assisted therapy session. The overall response and remission rates in this study were 83% and 58% (Ross et al., 2016). Similarly, a study by Griffiths and colleagues (2016) including anxious and/or depressed terminally ill patients (n=51) saw large dose-dependent decreases in depressive and anxious symptoms following two psilocybin experiences. This study did not include a guided therapy session, simply having participants lay down, wear an eye cover, and listen to music through a headset during treatment. Clinician-rated measures of depressive symptoms in this study showed response and remission rates of 78% and 65%, while measures of anxious symptoms showed response and remission rates of 83% and 57%. These decreases in symptoms were sustained for at least six months following this treatment by 80% of participants (Griffiths et al., 2016). Classic psychedelics rarely incur serious negative side effects, though nausea and vomiting, fatigue, psychological discomfort, and anxiety may be experienced during the acute phase (Griffiths et al., 2016). These adverse effects are dose-dependent and tend to wear off within 24 hours of ingestion (Studerus et al., 2011; Zentner, 1985). Additional positive long-term effects of psychedelics include improved convergent thinking/creativity, empathy, trait openness and cognitive and behavioural flexibility, lasting up to two years after a single dose (Agin-Liebes et al., 2020; Mans et. al., 2021; Mason et al., 2019). Clinical studies have also highlighted a link between the acute psychedelic experience and the lasting therapeutic effects.

For example, Roseman and colleagues (2018) found in an analysis of 20 participants' therapeutic outcomes from psilocybin treatment that the quality of mystical experience during treatment enhanced therapeutic outcomes, while anxiety and negative expectations negatively influenced outcomes.

Mechanisms of Effect

Ingesting LSD or psilocybin results in dose-dependent alterations in waking consciousness through profound changes of mood, perception, thought and experience of self, lasting 12-16 hours for LSD, and two to four hours for psilocybin (Schmid et al., 2021; Studerus et al., 2011). Ingesting psychedelics is thought to enhance neural plasticity, inter-network communication within the brain and autobiographical memory (Carhart-Harris et al., 2012a; Roseman et al., 2014; Nichols et al., 2017). These effects have been described as enabling "a state of unconstrained cognition" (Carhart-Harris et al., 2012 p. 1). This subjective experience of psychedelics appears to be linked directly with their agonistic effect on the serotonin 2A receptors in the brain. This has been demonstrated by the ability of 5-HT2A receptor antagonists such as ketanserin's to block the acute psychedelic effects (Glennon et al., 1984; Vollenweider et al., 1998). These 5-HT2A receptors are located in the cerebral cortex, most densely in regions associated with cognitive functions such as perception, thought, memory and consciousness. A small study (n= 12) by Barrett and colleagues (2020) used fMRI and emotion processing tasks to investigate possible sustained changes in brain function and affect following a single psilocybin dose. Their results showed decreases in negative affect, decreased amygdala response to emotional stimuli and increased activity in reward-learning, attention, and decision-making brain circuitry one week following treatment. Global increases in functional connectivity within the brain, which were sustained one month after treatment were also reported (Barrett et al., 2020).

fMRI studies have also shown a decrease in activity in the medial prefrontal cortex (mPFC) and posterior cingulate cortex (PCC) by up to 20% during the psychedelic state (Carhart-Harris et al., 2012; Carhart-Harris et al., 2012a; Muthukumaraswamy et al., 2013). The mPFC and PCC are both associated with the Default Mode Network (DMN), so reduced activity in these two regions causes reduced connectivity within the DMN (Muthukumaraswamy et al., 2013). The DMN is usually more active while the individual is resting rather than completing an externally focused task, and supports activities such as introspection, theory-of-mind, and mental time-travel (Spreng & Grady, 2010; Buckner & Carroll, 2007). Interestingly, increased activity and connectivity between the mPFC and other brain regions, as well as increased DMN functional connectivity has been associated with depressive rumination and trait neuroticism in past studies (Sheline et al., 2010; Berman et al., 2011; Adelstein; Farb et al., 2011). Following the psychedelic state, increased connectivity between the task positive network (TPN) and DMN has been observed (Carhart-Harris, 2014), which can also be seen in some meditative states and is thought to represent the two separate concepts of internal and external states narrowing (Froeliger et al., 2012; Carhart-Harris et al., 2012). Through destabilising well-established networks of the brain and narrowing the degree of separation between them, it is hypothesised that psychedelics have such lasting positive effects on affect and cognition by providing an opportunity for the resetting of neural pathways, which has a lasting effect following the resolution of the acute psychedelic effects (Nichols, 2017).

Preclinical Studies

Preclinical animal models have not yet been fully utilised in this area of research, though they are likely to be useful as they enable proper blinding and placebo control. In addition, they allow for a detailed examination of the underlying neurobiological mechanisms. The few past animal studies on the psychedelic treatment of psychological disorders have returned mixed results. Jefsen and colleagues (2019) found no therapeutic effect of psilocybin in their study using the Flinders Sensitive Line (FSL) genetic model of depression, as measured by the Porsolt forced swim test (FST) of depression (Porsolt et al., 1977). FSL rats are a widely validated and commonly used genetic model of depression, however the use of this model in studying the therapeutic effects of psychedelic substances has been criticised, as these rats exhibit significantly decreased mRNA expression of 5-HT2A. Therefore, the therapeutic potential of psychedelics is likely limited in FSL rats, as their main mechanism of effect seems to be via the 5HT2A receptors (Hibicke et al., 2020). Hibicke and colleagues conducted a study in 2020, instead using Wistar-Kyoto (WKY) rats, who exhibit similar vulnerabilities to stress as FSL rats, and are a well validated model of depression, but do not exhibit decreased 5-HT2A function. This study reported significant decreases in depressive symptoms as measured by the FST, and significant decreases in anxious symptoms as measured by the elevated plus maze test (EPM), five weeks after an injection of 1 mg/kg of psilocybin ($d = 4.985$). A similar, though weaker effect was also reported following a single injection of 0.15 mg/kg of LSD ($d = 0.863$) (Hibicke et al., 2020). The FST and EPM are not, however, the most appropriate tests for this context. The FST is now generally regarded as an animal model of coping strategy, rather than of depression-like symptoms, and the EPM suffers from several shortcomings. These include ambiguous analysis of behaviours on the EPM, a one-trial tolerance, and the lack of incentive to explore the open arms, particularly upon repeated exposure. These issues are discussed in further detail elsewhere (File et al., 1990; Molendijk et al., 2015; Pereira et al., 1999; Rodgers et al., 1995). Additionally, these studies were conducted solely in males. While common in animal research, this approach is questionable, given the clear sex/gender differences in human affective disorders.

Sex Differences

Sex differences in the prevalence and symptomatology of affective disorders have been well-established within humans, with women suffering from twice the lifetime risk of depression and most anxiety disorders than do men (Kessler et al., 1994; Weissman et al., 1994). It is unclear whether this is due to inherent biological differences, or social phenomena resulting in increased diagnoses for women, or a combination of both. Some have suggested that female hormonal, psychological and behavioural features may be related to increased susceptibility to affective disorders (Taylor et al., 2000; Martel, 2013). For example, superior social cognition and improved language abilities seen in women may also increase sensitivity to separation, criticism, and rejection, which are pronounced features of depression and anxiety (Altemus et al., 2014). However, many of these sex differences are not observed until young adulthood, by which time it is difficult to separate biological and social factors (Altemus et al., 2014). No sex differences in the acute or therapeutic effects of psychedelic substances have been observed in human studies to date, though some have reported a reduced acute behavioural response to psychedelics in female rats (Tylš et al., 2016; Páleníček et al., 2010). This acute sex difference may be relevant to the therapeutic effects of psychedelics on rats, considering the Roseman et al. (2018) finding that the acute psychedelic experience appears to influence the subsequent therapeutic outcomes.

The Current Experiment

To further the preclinical research, we used an early maternal separation (EMS) model of depression to test the therapeutic effects of psilocybin, using a novel test developed in the Victoria University of Wellington Behavioural Neurogenetics Lab, called the Affective Disorders Test (ADT). We tested the acute and sub-acute therapeutic effects of a single dose of psilocybin on anxious and depressive symptoms in male and female EMS rats, additionally testing for any sex differences in these effects. Based on past clinical research, we predicted first that subjects would exhibit significant decreases in depressive and anxious symptoms following psilocybin treatment compared with placebo. Second, based on the Tylš et al. finding, our second prediction was that female subjects would exhibit decreased behavioural effects during the psilocybin treatment than males. Finally, due to past findings on the acute psychedelic experience's influence on therapeutic outcomes, we predicted that the females would subsequently exhibit a decreased therapeutic improvement following treatment, compared with the males. Given the limited time, and the disruption induced by the Covid-19 pandemic, only a small number of animals in each group could be analysed. Hence the study represents a pilot analysis.

Methods

Animals

Outbred male and female wild type Sprague-Dawley rats were used in this study. Pups were housed with their mother until postnatal day (PND) 21, when pups were weaned and moved to new housing cages in pairs, separated by sex. These 485 x 340 x 213 mm (length x width x height) housing cages were kept under controlled temperature (21°C), lighting (12h light, 12h dark cycle) and humidity (55%) conditions throughout the experiment. This study was approved by the Animal Ethics Committee of Victoria University of Wellington.

On the day of birth (PND 0), subjects were randomly allocated to experimental and treatment conditions, detailed below (Table 1). EMS commenced on PND four and continued until PND 14 for subjects allocated to this condition. The Affective Disorders Test (ADT) was selected to measure anxious and depressive symptoms. This test requires that the subjects' food intake is restricted, which was done from PND 50 onward. Affective disorders testing commenced on PND 60 for five days, following which the psilocybin treatment was administered via injection on postnatal day 65. Following treatment, subjects underwent a further three days of behavioural testing on PNDs 66-68.

Table 1

Subject Groups

	Treatment Condition			
	Males (n = 17)		**Females (n = 17)**	
Experimental Condition	EMS / Saline n = 2	Control / Saline n = 4	EMS / Saline n = 2	Control / Saline n = 3
	EMS / 8 mg/kg Psilocybin n = 3	Control / 8 mg/kg Psilocybin n = 3	EMS / 8 mg/kg Psilocybin n = 3	Control /8 mg/kg Psilocybin n = 3
	EMS / 16 mg/kg Psilocybin n = 2	Control / 16 mg/kg Psilocybin n = 3	EMS /16 mg/kg Psilocybin n = 2	Control / 16 mg/kg Psilocybin n = 3

Early Maternal Separation (EMS)

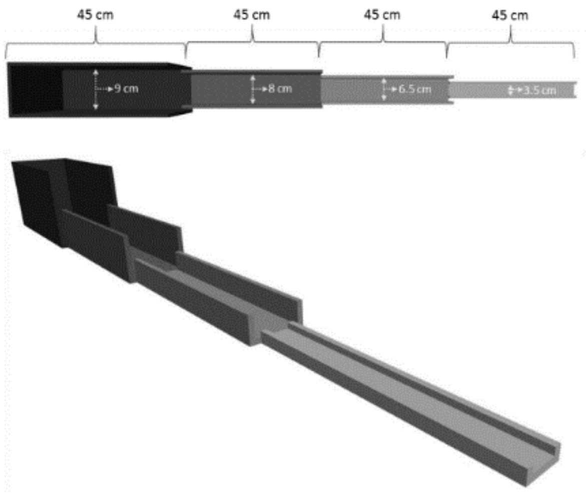
The method to induce signs of affective disorders was EMS. Pups allocated to the EMS condition were separated from their mother (dam) for three hours per day, on 10 consecutive days. This occurred between PNDs four and 14. For these three hours of separation each day, the dam was moved to a separate housing cage and relocated to a different room, causing significant stress for the dam and her pups. On each of the 10 days, the dams of control litters were handled for one minute, then returned to their pups, to create a consistent control condition. Past studies have shown that rat pups exposed to stress of this kind exhibit increased anxious- and depressive-like behaviours in adulthood (Hall, 1998; Lee et al., 2007). This link is theorised to be mediated by increased activity in the hypothalamic pituitary adrenal (HPA) axis, resulting in decreased gene expression of brain derived neurotrophic factor (BDNF) (Aisa et al., 2007), and decreased reuptake of 5-HT (Lee et al., 2007). This long-term hyperactivation of the HPA axis following early life stress also happens in humans (Heim et al., 2008; Juruena et al., 2020), resulting in increased cortisol levels and decreased BDNF, which have been reported in human patients with depression (Bhagwagar et al., 2005; Martinowich et al., 2007). The behavioural changes in adulthood resulting from the EMS paradigm have also been shown to improve following escitalopram (an SSRI) treatment, establishing the paradigm's predictive validity (El Khoury et al., 2006; do Couto et al., 2012).

Affective Disorders Test (ADT)

The ADT, consisting of the Anticipatory Pleasure (AP) Paradigm, and the Successive Alleys Test (SAT), was chosen to measure subjects' anxious and depressive symptoms in this study. The SAT was introduced as an alternative to the EPM test in 2013 by Deacon. The SAT consists of four successive elevated alleys with increasing anxiogenic characteristics (see figure 1). Alley one is the darkest, widest alley with the highest walls, and therefore feels the safest to the subjects. Alley four is the most anxiogenic alley, where it is brighter, more exposed, and thinner. This test was modified in 2019 to include a desirable bait at the end of each alley (half a Froot Loop), acting as an incentive for exploration of the area (Kidwell, 2019). This use of a reward makes the test effective over many trials, incentivising subjects to traverse the anxiogenic alleys after they lose their novelty. Subjects' food intake was restricted to ensure sufficient enticement, with the aim of subjects retaining 95% of their free feeding weight.

Figure 1

Image of the Successive Alleys Test



The addition of an anticipatory arena (40 x 40 x 40 cm) further increases the utility of this test, making it possible to measure anxious-like and depressive-like behaviours within the same test. Together, the modified SAT and the anticipatory arena make up the novel affective disorders test, developed in the Victoria University of Wellington Behavioural Neurogenetics Lab. Subjects spend five minutes in this anticipatory arena preceding each SAT trial, resulting in the pairing of this area with the shortly following access to desirable food rewards.

ADT Measures

Behaviour on the SAT was measured to reflect anxious symptoms. Time spent in alley 4 was measured in seconds, with the expectation that EMS subjects would spend less time in this area. Latency to eat the fourth reward was measured in seconds, with the expectation that EMS subjects would be more hesitant and therefore take longer to eat the fourth reward. Third, the locations of reward consumption were measured, with the expectation that EMS subjects would prefer to collect their reward and carry it back to the safer alleys for consumption. Less anxious rats, on the other hand, are expected to exhibit less caution in this regard. This measure is recorded as a consumption score representing points gained for each reward. A reward eaten in alley one earns one point, while a reward eaten in alley two earns two points, and so on. There is a maximum possible consumption score of 16, with a higher score representing less anxious behaviour. These measures were scored manually, using video recordings.

Behavioural expression of anticipatory pleasure is expected from healthy rodents in the anticipatory arena, manifested through 50kHz ultrasonic vocalizations (USVs) and rearing on the hind legs. These behaviours are reported to express positive affect (Barbano & Cador, 2006; Buck et al., 2014), so by measuring them we can infer the amount of anticipatory pleasure the subjects experienced. Subjects who express less pleasure in this arena are theorised to be experiencing a similar symptom to anhedonia (Bevins & Besheer, 2005). Rearing was manually scored from video recordings, and USVs, which were recorded on an UltraMic microphone, were analysed using the software program DeepSqueak (an add-on to Matlab, 2021).

Treatment

Subjects were randomly assigned to one of four treatment groups. Group one received a single injection of 8 mg/kg of psilocybin; group two received a single intraperitoneal injection of 16 mg/kg of psilocybin; group three received a single saline injection matched in volume with the low dose psilocybin group; and the fourth group received a single saline injection matched in volume with the high dose psilocybin group. Prior to injection, subjects were placed in a novel arena (40 x 40 x 40 cm) for 10 minutes of habituation. Upon administration of treatment, subjects were placed back in the same arena, where their behaviour was observed and recorded for 30 minutes. After 30 minutes, subjects were returned to their housing cages.

Acute Measures

Several behavioural measures were scored to represent subjects' acute psilocybin experience. Rearing behaviour was again measured, as this is a common behaviour for rats in a novel environment. Time spent in a flat body posture was measured, as this is a common behaviour, representing the strength of the psilocybin effect. Past literature has suggested this behaviour is linked with sudden increases of serotonin in the brain (Haberzettl et al., 2013). These first two measures were scored manually using video recordings. Distance moved and velocity were also measured using a tracking software called Ethovision. All acute measures were recorded as duration (in seconds), except rearing, which was counted as numbers. We expected female subjects to rear more, spend less time in a flat body posture, and move further and faster than male subjects, representing a decreased psilocybin effect.

Statistical Analyses

First, we conducted an independent samples t-test to ensure that no significant differences could be detected between the low-volume and high-volume saline dose groups. No significant differences were found, so both saline groups were treated as one for all subsequent analyses.

Acute Analyses

Four ANOVA tests were conducted for the acute effects, using number of rears, length of time spent in flat body posture, distance moved, and velocity of movement as the outcome variables. The fixed factors used in all four of these tests were sex, which had two levels (male and female); condition, which had two levels (EMS and control); and dose, which had three levels (saline, 8mg/kg psilocybin, and 16mg/kg psilocybin).

ADT Analyses

Four Mixed Measures ANOVAs were conducted using SAT consumption score, latency to eat the fourth reward, time spent in alley four, and number of rears in the anticipatory arena as the outcome variables. These tests included three between subjects' factors which were sex, condition, and dose, as described above. Trial was used in each of these tests as the only repeated measures factor, which had two levels. These were trials five and six, which correspond to the final trial of baseline tests prior to treatment, and the first trial of the post-treatment ADTs, taking place one day following treatment.

Results

In this section, any significant effects found in our analyses will be described. Additionally, as the number of subjects used in this experiment was small, some almost significant trends in the data will also be discussed. These trends may warrant further investigation in a larger sample, as our study may not have been statistically powerful enough to detect a statistically significant effect. Table 2 offers an overview of the results, which are described in further detail below.

Missing Data

Unfortunately, due to technical issues with the microphone used to record the females in the anticipatory arena, we were unable to analyse the females' USVs. We analysed the males' USVs alone, via a Repeated Measures ANOVA, using trial as the repeated measure with two levels (trial five and six), and dose and condition as between subjects' factors. Additionally, data collected from one female subject in the control/saline group was removed early from the treatment arena due to several disruptive escape attempts during. This subject's data was not included in analyses.

Table 2

Overview of Significant and Almost Significant Results

Acute Data	**Measure**	**Effect/Interaction**	**p-value**	**η_2**
	Flat Body Posture	Dose	>.001	.560
		Dose x Condition	.056	.087
	Rears	Dose	>.001	.446
	Distance	Dose	.077	.167
	Velocity	Dose	.019	.265
ADT Data	Latency to Eat Fourth Reward	Trial	.005	.016
		Sex	.042	.023
		Condition	.081	.016
		Dose	.080	.028
		Trial x Condition	.095	.005
		Sex x Dose	.081	.027
		Sex x Condition	.003	.057
		Trial x Sex x Condition	.051	.007
	SAT score	Condition	.076	.018
		Trial x Condition x Dose	.038	.019
		Sex x Condition x Dose	.091	.027
	Time Spent in Alley 4	Sex	.024	.027
		Sex x Dose	.036	.036

	Trial	.014	.006
Male USVs	Trial x Dose	.014	.009
	Sex	<.001	.081
	Dose x Sex	.074	.028
Anticipatory Rears	Dose x Condition x Sex	.023	.044

Flat Body Posture

Dose Effect

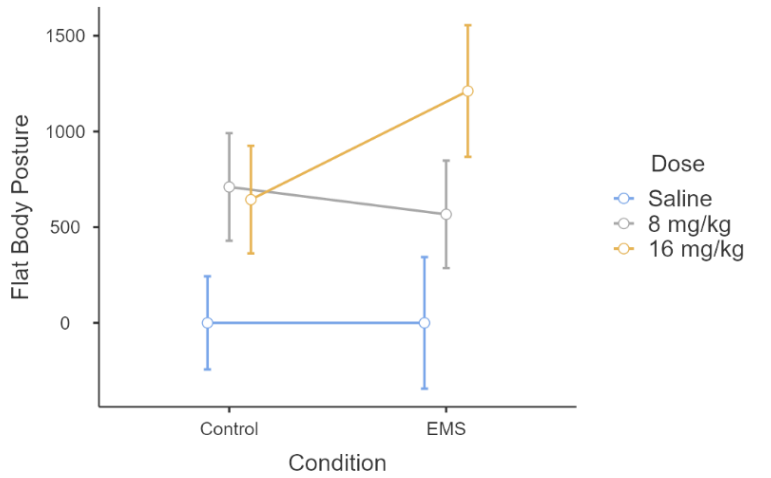
There was a main effect of dose on flat body posture (F = 21.13, df = 2, p >.001), in which time spent in flat body posture tended to increase with the psilocybin dose.

Dose x Condition Interaction

An almost significant (p = .056) interactive effect of dose by condition was also present in the flat body measure. Figure 2 demonstrates this interaction, showing that subjects from the EMS condition that received 16 mg/kg of psilocybin tended to spend longer in flat body posture than control subjects who received the same dose. This suggests that the high dose of psilocybin had a stronger effect on the EMS group than controls.

Figure 2

Line Graph Demonstrating an Almost Significant Dose x Condition Interaction in the Flat Body Measure



Rears (Acute)

Dose Effect

A main effect of dose was found in rearing behaviours during treatment (F = 9.84, df = 2, p >.001), such that rearing significantly decreased if the subject received either dose of psilocybin.

Velocity

Dose Effect

A main effect of dose was also found on average velocity of movement during treatment (F = 4.78, df = 2, p = .019), with the saline group moving significantly faster than either of the psilocybin dose groups.

Distance

Dose Trend

An almost significant trend of dose was detected on average distance travelled during treatment (p = .077), where the saline dose group tended to travel further than the psilocybin dose groups.

Latency to Eat the Fourth Reward

Trial Effect

A significant main effect of trial was detected in the latency to eat the fourth reward measure (F = 9.66, df = 1, p = .005). This effect showed a significant decrease in latency from trial five to trial six, suggesting a decrease in anxiety.

Sex Effect

A significant main effect of sex was detected on latency to eat the fourth reward (F = 4.64, df = 1, p = .042), demonstrating that overall, our female subjects tended to take longer to reach and eat their fourth reward compared with the males. This suggests a more anxious tendency in the females than the males.

Condition Trend

Third, an almost significant condition effect was detected in the same measure (p = .081), such that overall, EMS subjects tended to take longer to reach and eat their fourth reward, displaying more anxious behaviour in this condition group.

Dose Trend

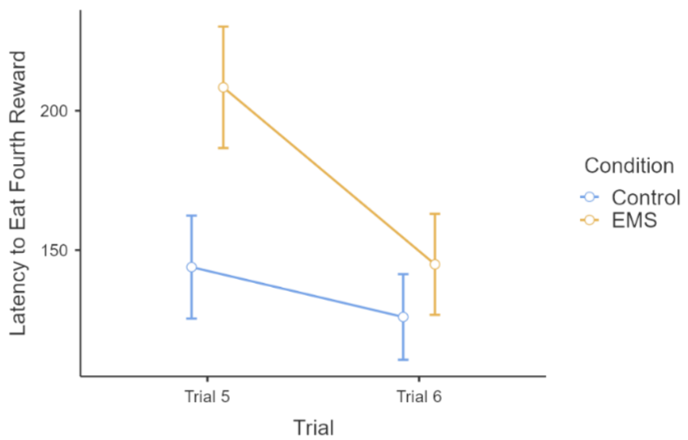
An almost significant main effect of dose was also found on latency to eat the fourth reward (p = .080), such that the 8 mg/kg dose group tended to take the longest to reach and eat their fourth reward, followed by the saline group, then the 16 mg/kg group, who were the fastest.

Trial x Condition Trend

An almost significant interactive trend was detected between experimental condition and trial on latency to eat the fourth reward (p = .095). As is displayed in figure 3, this interactive trend demonstrates that the difference between control and EMS subjects' time taken to eat the fourth reward narrowed significantly following treatment, due to a significant decrease in EMS scores.

Figure 3

Line Graph Demonstrating an Almost Significant Trial x Condition on Latency to Eat the Fourth Reward

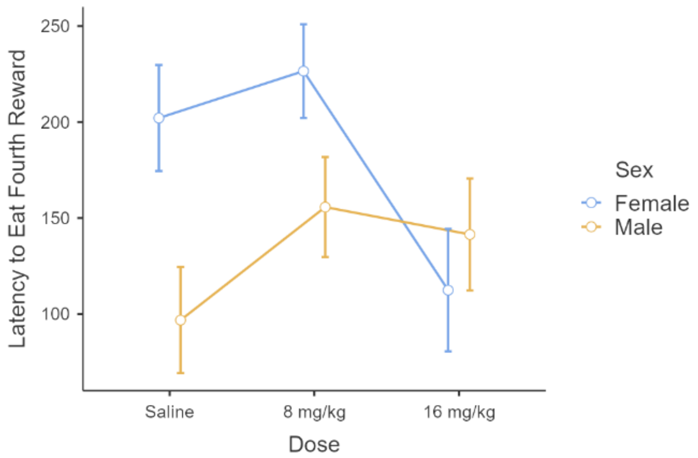


Sex x Dose Trend

An almost significant interactive trend between sex and dose was also detected in the same measure (*p* = .081). As is shown in figure 4, females in the saline and 8 mg/kg dose groups took longer to eat the fourth reward than females in the 16 mg/kg dose group., suggesting a therapeutic effect of the 16 mg/kg dose on females. Males in either psilocybin dose group took slightly longer to reach and eat the fourth than the males that received saline.

Figure 4

Line Graph Demonstrating the Almost Significant Sex by Dose Trend in Latency to Eat Fourth Reward Measure

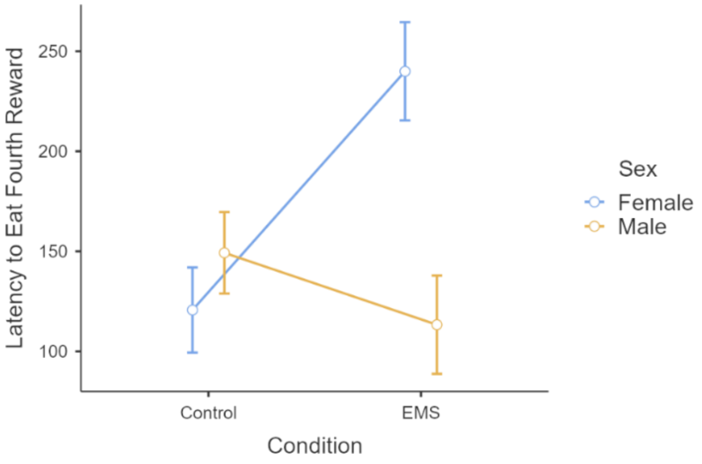


Sex x Condition Interaction

A significant interaction between sex and condition was also detected in the latency to eat the fourth reward measure (F = 11.62, df = 1 p = .003), as is displayed in figure 5. This interaction shows that females in the EMS condition took longer to reach and eat their fourth reward than control females, or males of either condition group. This suggests that EMS had a stronger anxiogenic effect on female subjects than male subjects.

Figure 5

Line Graph Demonstrating the Interaction on Latency to Eat the Fourth Reward

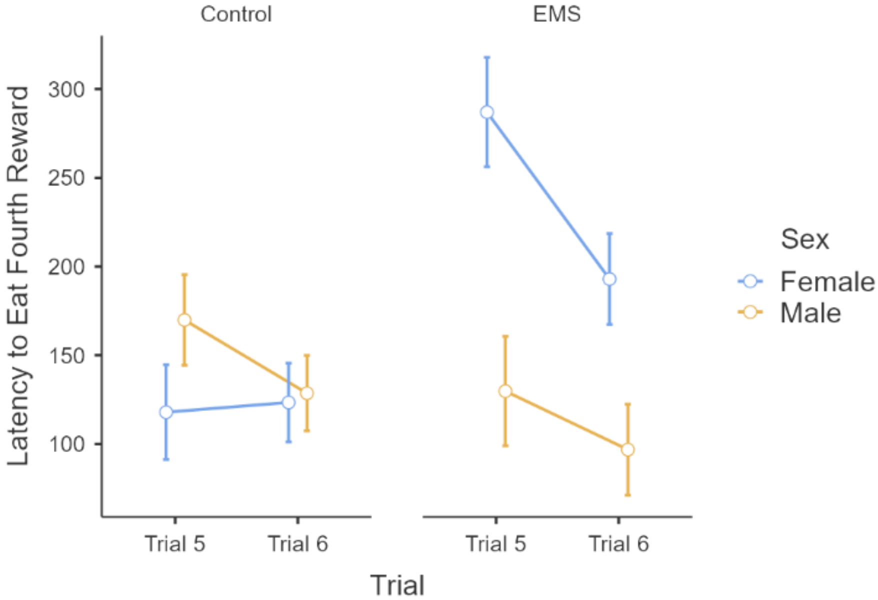


Trial x Sex x Condition Interactive Trend

The final finding for this measure was an almost significant trend of interaction between trial, sex, and condition (p = .051). EMS females took significantly less time to eat the fourth reward on trial six compared with trial five. This suggests a stronger therapeutic effect of the treatment on EMS female subjects than control females or males of either condition.

Figure 6

Line Graph Demonstrating the Almost Significant Trial x Sex x Condition Trend in Latency to Eat Fourth Reward Measure



SAT Consumption Scores

Condition Trend

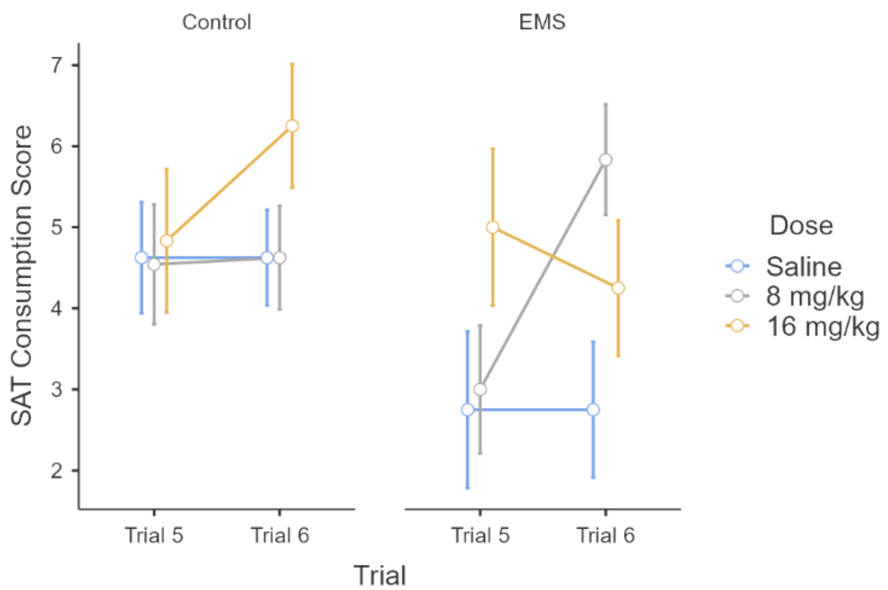
Moving on to the findings from the next ADT measure, an almost significant main effect of condition was detected in subjects' consumption scores (p = .076), displaying more anxious tendencies in the EMS group on this measure.

Trial x Condition x Dose Interaction

A significant interaction between trial, experimental condition and psilocybin dose was also found in the SAT consumption scores (F = 3.80, df = 2, p = .038), with the largest increase seen in the EMS subjects from the 8 mg/kg dose group. A small decrease in consumption score was seen in EMS subjects who received 16 mg/kg of psilocybin, suggesting a larger therapeutic effect of the 8 mg/kg dose than the 16 mg/kg dose of psilocybin across all EMS subjects. Interestingly, those from the control group who received 16 mg/kg of psilocybin showed an increase in consumption score, while those in the EMS condition who received 16 mg/kg of psilocybin showed a slight decrease in consumption score following treatment.

Figure 5

Line Graph Demonstrating a Trial x Condition x Dose Interaction on SAT Consumption Score

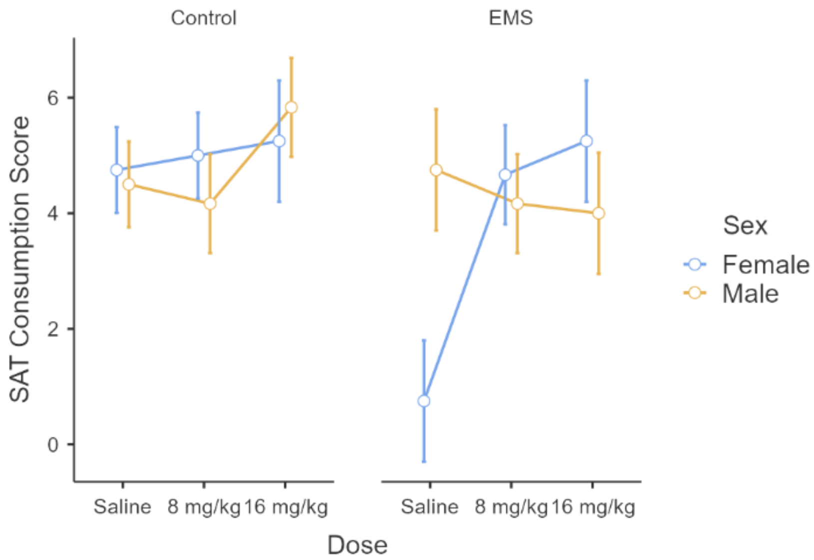


Sex x Dose x Condition Interactive Trend

An almost significant interactive trend between sex, dose, and condition was also found in the SAT consumption scores (p = .091), such that EMS females in both psilocybin dose groups showed less anxious consumption behaviours than EMS females than received saline. This trend is displayed below in figure 6.

Figure 6

Line Graph Demonstrating the Non-significant Sex x Dose x Condition Trend in SAT Consumption Scores



Time Spent in Alley Four

Sex Effect

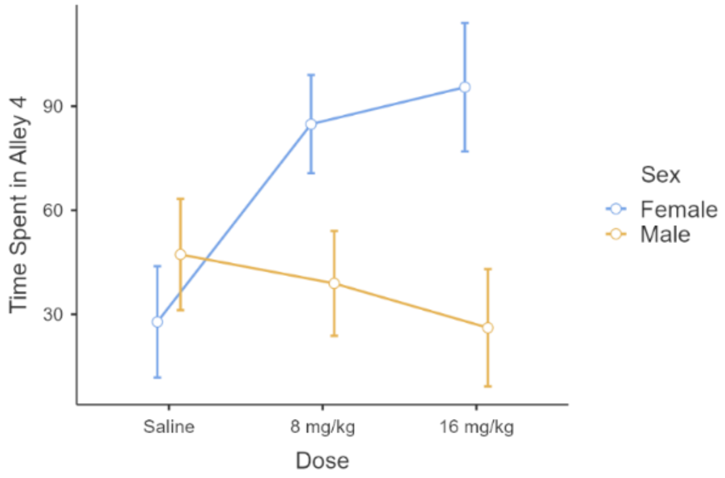
First, a significant main effect of sex was found in the time spend in alley 4 measure (F = 7.83, df = 1, p = .011), such that the males' scores were lower.

Sex x Dose Interaction

A significant interaction between sex and dose was found in the time spent in alley four scores (F = 4.76, df = 2, p = .020). As is displayed in figure 7, females in the 8mg/kg and 16 mg/kg groups displayed increased willingness to spend time in alley four, compared with females in the saline condition, suggesting decreased anxiety in the female psilocybin groups.

Figure 7

Line Graph Demonstrating the Sex x Dose Interaction on Time Spent in Alley 4



Anticipatory Rears

Sex Effect

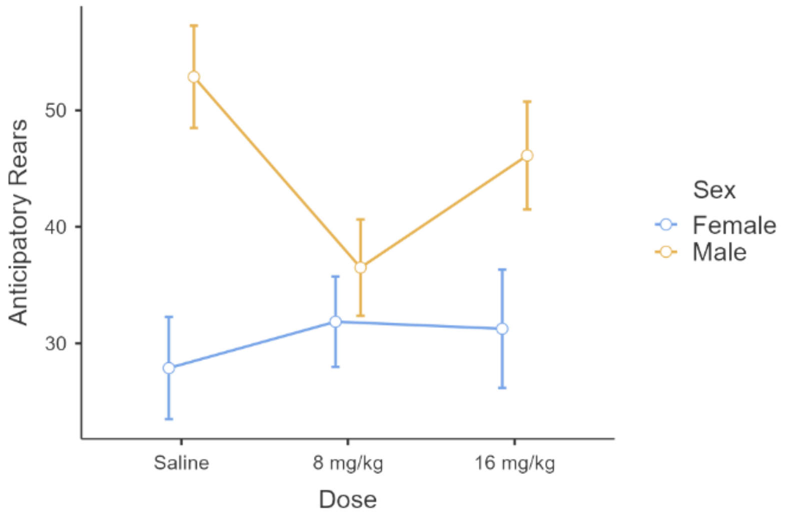
A main effect of sex was found on the number of rears subjects did in the anticipatory arena (F = 16.83, df = 1, p < .001). This effect reflected a tendency for females to rear less than males, suggesting overall more anhedonic behaviours in the female subjects.

Dose x Sex Interactive Trend

An almost significant interactive trend of dose and sex was found in the anticipatory rears measure (p = .074). This is displayed in figure 9. Females across all dose groups seemed to rear similarly often, but rears by males in the psilocybin groups were decreased compared with males that received saline.

Figure 9

Line Graph Demonstrating an Almost Significant Sex by Dose Interactive Trend on Anticipatory Rears

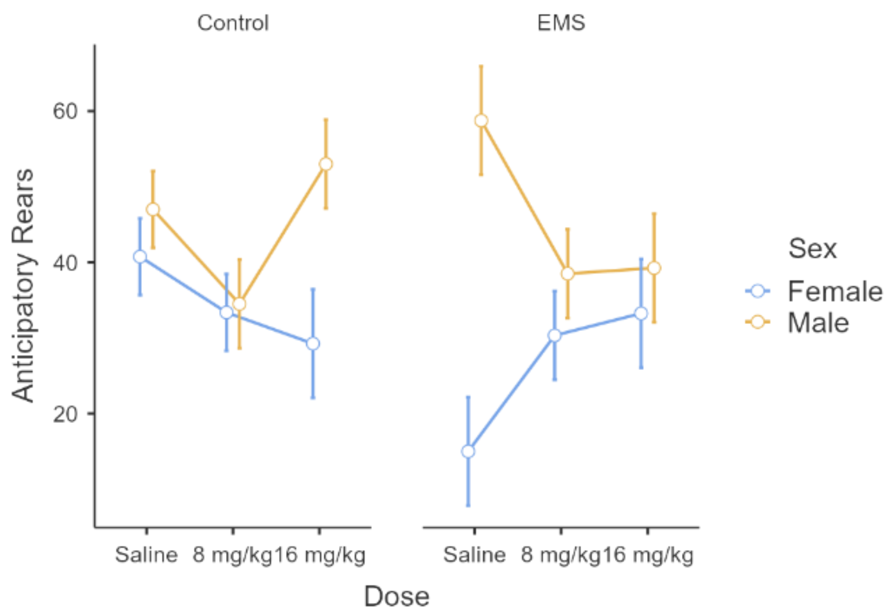


Dose x Condition x Sex Interaction

There was also a significant interaction between sex, dose, and condition on anticipatory rearing behaviour (F = 4.53, df = 2, p = .023). As is seen in figure 10, The EMS females in the 8mg/kg and 16 mg/kg dose groups reared more than EMS females that received saline. EMS males showed the opposite trend, with those in the 8 mg/kg and 16 mg/kg groups reareing less than EMS males in the saline treatment group.

Figure 10

Line Graph Demonstrating the Dose x Condition x Sex Interaction on Anticipatory Rears



Male USVs

Trial Effect

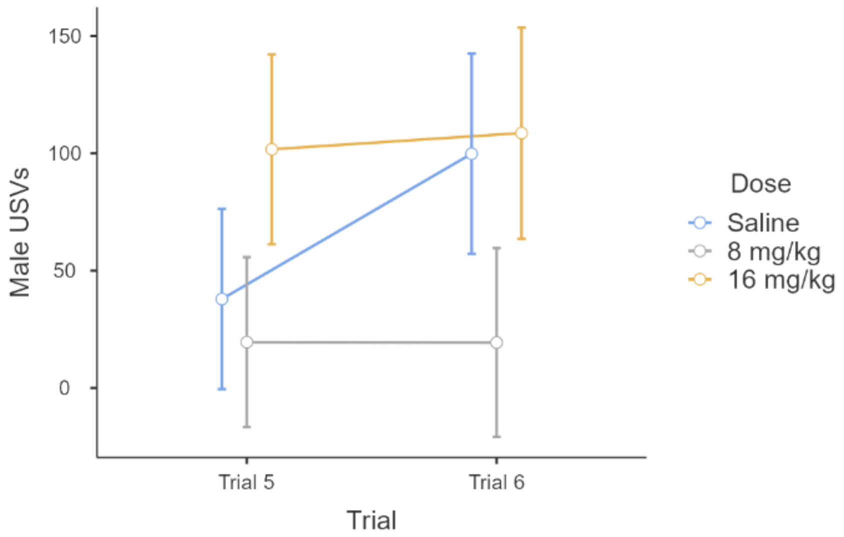
A significant main effect of trial was detected in the male USV data (F = 8.53, df = 1, p = .014). This effect reflects a tendency for the males' USVs to increase from trial five to six, indicating an increase in anticipatory pleasure.

Trial x Dose Interaction

Finally, a significant interaction of trial with dose was also found on anticipatory rears (F = 6.38, df = 2, p = .014), which represents the a increase across trials in USVs by males in the saline group, but no increase in USVs by males in either psilocybin dose group.

Figure 11

Line Graph Demonstrating the Trial x Dose Interaction in Male USVs



Discussion

No Acute Sex Effect

The first main finding of this experiment is a lack of sex effect in the acute measures. This finding does not support hypothesis 2a, or the Tylš et al. (2016) finding of a decreased acute effect of psilocybin in females compared with males. There are, however, several possible reasons we failed to detect a sex effect. First, we exclusively analysed subjects' naturally occurring behaviours during the acute stage, while Tylš et al. conducted pre-pulse inhibition (PPI) tests and open field (OP) tests during the acute phase, which may have given a clearer view of the internal acute experience (Tylš et al., 2016; Páleníček et al., 2010). PPI was also used during the acute period of psilocybin treatment by Miliano and colleagues (2019), which demonstrated a sex effect supporting the Tylš et al. finding, lending support to this method.

Second, in this study we neglected to measure and control for the females' oestrus phase, as was done by Tylš et al. In their study, the decreased effect of psilocybin was most pronounced in females in the pre-oestrus and oestrus phases (Tylš et al., 2016). Thus, it is possible that neglecting to separate female subjects by menstrual phase decreased our ability to detect this sex effect, which may be weaker during the metestrus and dioestrus phases.

A third possibility is that the sex effect found by Tylš et al. is dose dependent and becomes less pronounced with higher doses. The current study used significantly higher doses than those used in the Tylš et al. study, who administered only 1, 2, and 4 mg/kg of psilocin.

Thus, although this experiment was unable to detect a sex effect in acute psilocybin experience, this null finding may lend support to the existence of a small oestrus phase-specific, dose-dependent sex effect, which the PPI test may be most adept in detecting as opposed to naturally occurring behavioural measures. Further studies in which these specific conditions are met will be required to confirm or disconfirm this possibility.

Sex Effect of Therapeutic Outcome

The second main finding of this experiment is a therapeutic effect of psilocybin, demonstrated by the trial effects found in the male USVs and latency to eat the fourth reward data. These effects demonstrated that latency to eat the fourth reward decreased, and male USVs increased from pre-treatment to post-treatment. Interestingly, the therapeutic effect was driven mainly by females' scores, which is demonstrated by the interactions between sex and dose seen in the time spent in alley four and latency to eat the fourth reward measures. This finding partially supports hypothesis 1 and does not support hypothesis 2b.

One possible reason for this finding is the much stronger effect of EMS in females, as is demonstrated by the interaction found between sex and condition in the latency to eat the fourth reward measure. This increase in anxious and depressive behaviours in response to EMS may have offered more of an opportunity for a therapeutic effect in the females, while a floor effect may have been present for the male subjects. This possibility does not appear to be supported by the male USVs however, which demonstrate an increase in 50kHz USVs following treatment by the male saline group only.

An alternative is that the stress of the treatment may have influenced subsequent therapeutic effects, and that the females experienced less stress during treatment than the males. As was mentioned in our introduction, stress/anxiety levels appear to affect the therapeutic outcomes of psychedelic treatment (Roseman, 2018). Therefore, we suggest that the method of treatment in this experiment may have caused significant stress due to the high doses used and subjects' lack of familiarity with the substance. The interaction of sex, dose, and condition in anticipatory rears support this possibility, showing an increase in anticipatory rears by EMS females in both psilocybin groups, but a decrease in rears by EMS males in both psilocybin groups. We suggest that this differed response may reflect a sex difference in acute experience which this experiment was not able to detect due to its statistical and methodological limitations. If females do in fact have a decreased acute effect, this could theoretically reduce stress during treatment, in turn enhancing therapeutic effects. To test this possibility, future preclinical studies should pay special attention to subjects' stress levels during treatment. This can be done by analysing USVs and measuring heart rate and hear rate variability during treatment. Subjects should make more 22kHz calls than 50kHz calls and exhibit increased heart rate, as well as increased low frequency and decreased high frequency heart rate variability when significantly stressed. Alternatively, stress could be manipulated by playing recordings of 50kHz calls during treatment and including environmental enrichment and food rewards, or by playing a recording of 22kHz calls during treatment and excluding environmental enrichment and food rewards. Stress levels could then be analysed for interactions with sex and their acute and therapeutic responses.

A third interpretation of the sex difference in therapeutic effect found in this study is that females react more strongly to serotonergic drugs due to neurological and hormonal differences. This idea is supported in a review by Sramek and colleagues (2016) on sex differences in serotonergic depression treatment responses seen in humans. Schneider et al. (1997) are cited in this review, who found that oestrogen supplementation in menopausal women improved SSRI response. This finding, among others, supports the hypothesis that oestrogen levels are relevant to serotonin functioning and positively influence serotonergic drug response (Lokuge et al., 2011; Payne, 2003). Considering this, our female subjects may have been more sensitive to the therapeutic effects of psilocybin due to their increased oestrogen levels compared with the males. Future studies should consider comparing the therapeutic effects of psilocybin in females based on oestrus phase to investigate this.

An additional interesting extension of this study would be an experiment testing the sex effect of psilocybin's therapeutic outcome found here on long term anxious and depressive symptoms. This study only analysed the ADT trials taking place one day before and one day after treatment, therefore only capturing short term therapeutic responses, so we cannot rule out the possibility of a delayed therapeutic effect in the male subjects and cannot speak to the length of time that these improvements in anxious and depressive symptoms in females persisted for.

Limitations

The findings of this study must be considered within its limitations, including its low statistical power due to the small number of subjects. In addition to the limitations mentioned above, time spent in alley 4 proved to be relatively ambiguous in the anxious behaviours it measured. This became clear from the sex effect found on this measure, which showed that our males spent less time overall in alley 4, supposedly representing a more anxious male group. This does not match the findings from our other measures, which consistently showed a tendency for the females to display more anxious and depressive behaviours. This was shown by the sex effect in latency to eat the fourth reward, and anticipatory rears. We suggest that in some cases, more time spent in alley 4 may reflect an anxious tendency to check the area and slowly retrieve the reward, rather than efficiently and confidently retrieving the reward without worry of the safety of the area. Due to this ambiguity, we suggest that latency to eat the fourth reward and consumption score may be more representative measures of anxiety on the SAT.

Finally, the flat body measure was relatively difficult to score consistently. This measure requires a significant amount of judgement by the scorer as to the point at which the rat’s body should be counted as flat. This was especially difficult to determine from a top-down view, so future studies should use a side-on view, and test for interrater reliability between multiple scorers.

Implications

The findings of this experiment have several important implications for research in this area. First, they offer support for a therapeutic effect of psilocybin in rats. This is an important contribution to the study of psychedelics, as it supports a mechanism of effect unique to the substance itself, without the requirement of psychotherapy. With further support in larger clinical and preclinical studies, this information may contribute to the development of psilocybin as an alternative, single-dose treatment for depression, without the requirement of psychotherapy, improving accessibility.

Our findings may also offer support for a sex effect in serotonergic depression treatment responses. This information is relevant to the treatment of depression using serotonergic antidepressants and may inform the selection of appropriate doses based on sex and oestrogen levels.

Conclusion

We conclude that psilocybin has a therapeutic effect on anxious and depressive symptoms in rats and suggest that female rats are more sensitive to this therapeutic effect than males. Future research is needed however, to confirm that this sex effect of therapeutic outcome was not due to the low baseline level of anxious and depressive symptoms in the males of our sample. Due to low statistical power and methodological limitations, we are unable to confirm a lack of sex effect in acute psilocybin response based on our finding. Further research is needed to confirm any influences oestral phase and dose might have on this effect.

References

Adelstein, J. S., Shehzad, Z., Mennes, M., DeYoung, C. G., Zuo, X. N., Kelly, C., ... & Milham, M. P. (2011). Personality is reflected in the brain's intrinsic functional architecture. *PloS one*, *6*(11), e27633.

Agin-Liebes, G. I., Malone, T., Yalch, M. M., Mennenga, S. E., Ponté, K. L., Guss, J., ... & Ross, S. (2020). Long-term follow-up of psilocybin-assisted psychotherapy for psychiatric and existential distress in patients with life-threatening cancer. *Journal of Psychopharmacology*, *34(*2), 155-166.

Aisa, B., Tordera, R., Lasheras, B., Del Río, J., & Ramírez, M. J. (2007). Cognitive impairment associated to HPA axis hyperactivity after maternal separation in rats. *Psychoneuroendocrinology*, *32*(3), 256-266.

Alenina, N., & Klempin, F. (2015). The role of serotonin in adult hippocampal neurogenesis. *Behavioural brain research*, *277*, 49-57.

Altemus, M., Sarvaiya, N., & Epperson, C. N. (2014). Sex differences in anxiety and depression clinical perspectives. *Frontiers in neuroendocrinology*, *35*(3), 320-330.

American Psychological Association. (2013). Diagnostic and statistical manual of mental disorders: DSM-5. (Fifth edition.).

American Psychiatric Association. (2014). Early treatment of anxiety disorders will save money in the workplace. *Mental Health Works*, 7-9.

Baldwin, D., Woods, R., Lawson, R., & Taylor, D. (2011). Efficacy of drug treatments for generalised anxiety disorder: systematic review and meta-analysis. *BMJ*, *342*(7798), 46–637. https://doi.org/10.1136/bmj.d1199

Bandelow, B., & Michaelis, S. (2015). Epidemiology of anxiety disorders in the 21st century. *Dialogues in clinical neuroscience, 17*(3), 327.

Barrett, F. S., Doss, M. K., Sepeda, N. D., Pekar, J. J., & Griffiths, R. R. (2020). Emotions and brain function are altered up to one month after a single high dose of psilocybin. *Scientific Reports*, *10*(1), 1-14.

Barbano, M. F., & Cador, M. (2006). Differential Regulation of the Consummatory, Motivational and Anticipatory Aspects of Feeding Behavior by Dopaminergic and Opioidergic Drugs. *Neuropsychopharmacology, 31*(7), 1371–1381. https://doi.org/10.1038/sj.npp.1300908

Berman, M. G., Peltier, S., Nee, D. E., Kross, E., Deldin, P. J., & Jonides, J. (2011). Depression, rumination and the default network. *Social cognitive and affective neuroscience, 6*(5), 548-555.

Bhagwagar, Z., Hafizi, S., & Cowen, P. J. (2005). Increased salivary cortisol after waking in depression. *Psychopharmacology*, *182*(1), 54-57.

Bouso, J. C., & Sánchez-Avilés, C. (2020). Traditional healing practices involving psychoactive plants and the global mental health agenda: opportunities, pitfalls, and challenges in the "right to science" framework. *Health and Human Rights*, *22*(1), 145.

Branchi, I., Santarelli, S., Capoccia, S., Poggini, S., D'Andrea, I., Cirulli, F., & Alleva, E. (2013). Antidepressant treatment outcome depends on the quality of the living environment: a pre-clinical investigation in mice. *PloS one*, *8*(4), e62226.

Bevins, R. A., & Besheer, J. (2005). Novelty reward as a measure of anhedonia. Neuroscience & Biobehavioral Reviews, 29(4-5), 707-714.

Buck, C. L., Vendruscolo, L. F., Koob, G. F., & George, O. (2014). Dopamine D1 and μ-opioid receptor antagonism blocks anticipatory 50 kHz ultrasonic vocalizations induced by palatable food cues in Wistar rats. *Psychopharmacology, 231*(5), 929–937. https://doi.org/10.1007/s00213-013-3307-2

Buckner, R. L., & Carroll, D. C. (2007). Self-projection and the brain. *Trends in cognitive sciences, 11*(2), 49-57.

Carhart-Harris, R. L., Bolstridge, M., Day, C. M. J., Rucker, J., Watts, R., Erritzoe, D. E., ... & Nutt, D. J. (2018). Psilocybin with psychological support for treatment-resistant depression: six-month follow-up. *Psychopharmacology*, *235*(2), 399-408.

Carhart-Harris, R. L., Erritzoe, D., Williams, T., Stone, J. M., Reed, L. J., Colasanti, A., ... & Nutt, D. J. (2012). Neural correlates of the psychedelic state as determined by fMRI studies with psilocybin. *Proceedings of the National Academy of Sciences*, *109*(6), 2138-2143.

Carhart-Harris, R. L., & Goodwin, G. M. (2017). The Therapeutic Potential of Psychedelic Drugs: Past, Present, and Future. *Neuropsychopharmacology, 42*(11), 2105–2113. https://doi.org/10.1038/npp.2017.84

Carhart-Harris, R. L., Leech, R., Hellyer, P. J., Shanahan, M., Feilding, A., Tagliazucchi, E., ... & Nutt, D. (2014). The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs. *Frontiers in human neuroscience, 8*, 20.

Carhart-Harris, R. L., Leech, R., Williams, T. M., Erritzoe, D., Abbasi, N., Bargiotas, T., ... & Nutt, D. J. (2012a). Implications for psychedelic-assisted psychotherapy: functional magnetic resonance imaging study with psilocybin. *The British Journal of Psychiatry, 200*(3), 238-244.

Cowen, P. J., & Browning, M. (2015). What has serotonin to do with depression? *World Psychiatry*, *14*(2), 158.

Cowen, P. J., & Sargent, P. A. (1997). Changes in plasma prolactin during SSRI treatment: evidence for a delayed increase in 5-HT neurotransmission. *Journal of psychopharmacology*, *11(*4), 345-348.

Cuijpers, P., Noma, H., Karyotaki, E., Vinkers, C. H., Cipriani, A., & Furukawa, T. A. (2020). A network meta-analysis of the effects of psychotherapies, pharmacotherapies and their combination in the treatment of adult depression. *World Psychiatry*, *19*(1), 92-107.

Cuijpers, P., van Straten, A., Warmerdam, L., & Andersson, G. (2009). Psychotherapy versus the combination of psychotherapy and pharmacotherapy in the treatment of depression: a meta-analysis. *Depression and anxiety*, *26*(3), 279-288.

Deacon, R. M. J. (2013). The successive alleys test of anxiety in mice and rats. Journal of Visualized Experiments : JoVE, (76). https://doi.org/10.3791/2705

Demyttenaere, K., & Jaspers, L. (2008). Bupropion and SSRI-induced side effects. *Journal of Psychopharmacology*, *22*(7), 792-804.

do Couto, F. S., Batalha, V. L., Valadas, J. S., Data-Franca, J., Ribeiro, J. A., & Lopes, L. V. (2012). Escitalopram improves memory deficits induced by maternal separation in the rat. *European journal of pharmacology*, *695*(1-3), 71-75.

El Khoury, A., Gruber, S. H., Mørk, A., & Mathé, A. A. (2006). Adult life behavioral consequences of early maternal separation are alleviated by escitalopram treatment in a rat model of depression. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, *30*(3), 535-540.

Farb, N. A., Anderson, A. K., Bloch, R. T., & Segal, Z. V. (2011). Mood-linked responses in medial prefrontal cortex predict relapse in patients with recurrent unipolar depression. *Biological psychiatry*, *70*(4), 366-372.

File, S. E., Mabbutt, P. S., & Hitchcott, P. K. (1990). Characterisation of the phenomenon of “one-trial tolerance” to the anxiolytic effect of chlordiazepoxide in the elevated plus-maze. *Psychopharmacology*, *102(*1), 98–101. https://doi.org/10.1007/BF02245751

Froeliger, B., Garland, E. L., Kozink, R. V., Modlin, L. A., Chen, N. K., McClernon, F. J., ... & Sobin, P. (2012). Meditation-state functional connectivity (msFC): strengthening of the dorsal attention network and beyond. *Evidence-Based Complementary and Alternative Medicine, 2012.*

Glennon, R. A., Titeler, M., & McKenney, J. D. (1984). Evidence for 5-HT2 involvement in the mechanism of action of hallucinogenic agents. *Life sciences*, *35*(25), 2505-2511.

Goldberg, S. B., Pace, B. T., Nicholas, C. R., Raison, C. L., & Hutson, P. R. (2020). The experimental effects of psilocybin on symptoms of anxiety and depression: A meta-analysis. *Psychiatry research*, *284*, 112749.

Gorman, J. M. (1996). Comorbid depression and anxiety spectrum disorders. *Depression and anxiety*, *4*(4), 160-168.

Griffiths, R. R., Johnson, M. W., Carducci, M. A., Umbricht, A., Richards, W. A., Richards, B. D., ... & Klinedinst, M. A. (2016). Psilocybin produces substantial and sustained decreases in depression and anxiety in patients with life-threatening cancer: A randomized double-blind trial. *Journal of psychopharmacology*, *30*(12), 1181-1197.

Guerra-Doce, E. (2015). Psychoactive substances in prehistoric times: examining the archaeological evidence. *Time and Mind*, *8*(1), 91-112.

Gutiérrez-Rojas, L., Porras-Segovia, A., Dunne, H., Andrade-González, N., & Cervilla, J. A. (2020). Prevalence and correlates of major depressive disorder: a systematic review. *Brazilian Journal of Psychiatry*, *42*, 657-672.

Haberzettl, R., Bert, B., Fink, H., & Fox, M. A. (2013). Animal models of the serotonin syndrome: a systematic review. *Behavioural brain research*, *256*, 328-345.

Hall, F. S. (1998). Social deprivation of neonatal, adolescent, and adult rats has distinct neurochemical and behavioral consequences. *Critical Reviews™ in Neurobiology*, *12*(1-2).

Hazlett-Stevens, H. and Craske, H. G. (2002). Brief cognitive-behavioral therapy: definition and scientific foundations. In Bond, F. W. and Dryden, W. (Eds.) *Handbook of brief cognitive behaviour therapy* (pp. 1-20).

Heim, C., Newport, D. J., Mletzko, T., Miller, A. H., & Nemeroff, C. B. (2008). The link between childhood trauma and depression: insights from HPA axis studies in humans. Psychoneuroendocrinology, 33(6), 693-710.

Hibicke, M., Landry, A. N., Kramer, H. M., Talman, Z. K., & Nichols, C. D. (2020). Psychedelics, but not ketamine, produce persistent antidepressant-like effects in a rodent experimental system for the study of depression. *ACS chemical neuroscience*, *11*(6), 864-871.

Hofmann, A. (1980). LSD: my problem child. Psychedelic Reflections, 24.

Hofmann, S. G., Curtiss, J., Carpenter, J. K., & Kind, S. (2017). Effect of treatments for depression on quality of life: a meta-analysis. *Cognitive behaviour therapy*, *46*(4), 265-286.

Jefsen, O., Højgaard, K., Christiansen, S. L., Elfving, B., Nutt, D. J., Wegener, G., & Müller, H. K. (2019). Psilocybin lacks antidepressant-like effect in the Flinders Sensitive Line rat. *Acta neuropsychiatrica*, *31*(4), 213-219.

Juruena, M. F., Eror, F., Cleare, A. J., & Young, A. H. (2020). The role of early life stress in HPA axis and anxiety. *Anxiety Disorders,* 141-153.

Kessler, R. C., McGonagle, K. A., Zhao, S., Nelson, C. B., Hughes, M., Eshleman, S., ... & Kendler, K. S. (1994). Lifetime and 12-month prevalence of DSM-III-R psychiatric disorders in the United States: results from the National Comorbidity Survey. *Archives of general psychiatry*, *51*(1), 8-19.

Kidwell, M. (2019). *Establishment of an Animal Model of Depression: The Serotonin Transporter Knockout Rat*. Victoria University of Wellington, Wellington.

Kupfer, D. J. (2005). The pharmacological management of depression. *Dialogues in clinical neuroscience*, *7*(3), 191.

Lader, M. (2008). Effectiveness of benzodiazepines: do they work or not? *Expert Review of Neurotherapeutics*, *8*(8), 1189-1191.

Lee, J. H., Kim, H. J., Kim, J. G., Ryu, V., Kim, B. T., Kang, D. W., & Jahng, J. W. (2007). Depressive behaviors and decreased expression of serotonin reuptake transporter in rats that experienced neonatal maternal separation. *Neuroscience research, 58*(1), 32-39.

Lokuge, S., Frey, B. N., Foster, J. A., Soares, C. N., & Steiner, M. (2011). Depression in women: windows of vulnerability and new insights into the link between estrogen and serotonin. *The Journal of clinical psychiatry, 72*(11), 0-0.

Luoma, J. B., Chwyl, C., Bathje, G. J., Davis, A. K., & Lancelotta, R. (2020). A meta-analysis of placebo-controlled trials of psychedelic-assisted therapy. *Journal of Psychoactive Drugs*, *52*(4), 289-299.

Malhi, G. S., Bassett, D., Boyce, P., Bryant, R., Fitzgerald, P. B., Fritz, K., ... & Singh, A. B. (2015). Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for mood disorders. *Australian & New Zealand Journal of Psychiatry*, *49*(12), 1087-1206.

Mans, K., Kettner, H., Erritzoe, D., Haijen, E. C., Kaelen, M., & Carhart-Harris, R. L. (2021). Sustained, multifaceted improvements in mental well-being following psychedelic experiences in a prospective opportunity sample. *Frontiers in psychiatry, 12*, 1038.

Martel, M. M. (2013). Sexual selection and sex differences in the prevalence of childhood externalizing and adolescent internalizing disorders. *Psychological bulletin*, *139*(6), 1221.

Martin, J. L. R., Sainz-Pardo, M., Furukawa, T. A., Martin-Sanchez, E., Seoane, T., & Galan, C. (2007). Review: Benzodiazepines in generalized anxiety disorder: heterogeneity of outcomes based on a systematic review and meta-analysis of clinical trials. *Journal of Psychopharmacology*, *21*(7), 774–782. https://doi.org/10.1177/0269881107077355

Martinowich, K., Manji, H., & Lu, B. (2007). New insights into BDNF function in depression and anxiety. *Nature neuroscience*, *10*(9), 1089-1093.

Mason, N. L., Mischler, E., Uthaug, M. V., & Kuypers, K. P. (2019). Sub-acute effects of psilocybin on empathy, creative thinking, and subjective well-being. *Journal of psychoactive drugs*, *51*(2), 123-134.

MATLAB. (2021). *Version 9.10.0 (R2021a).* Natick, Massachusetts: The MathWorks Inc.

Molendijk, M. L., & de Kloet, E. R. (2015). Immobility in the forced swim test is adaptive and does not reflect depression. *Psychoneuroendocrinology, 62*, 389-391.

Ministry of Health. (2021). *Data on antidepressant prescriptions 2017-2020.* Wellington, New Zealand: Gaynor Bradfield.

National Institute for Health and Care Excellence. (2015). First-choice antidepressant use in adults with depression or generalised anxiety disorder.

National Institute of Mental Health. (2018). Suicide. Retrieved from https://www.nimh.nih.gov/health/statistics/suicide on 25/06/2021.

Nichols, D. E., Johnson, M. W., & Nichols, C. D. (2017). Psychedelics as medicines: an emerging new paradigm. *Clinical Pharmacology & Therapeutics*, *101*(2), 209-219.

Nutt, D. J., Forshall, S., Bell, C., Rich, A., Sandford, J., Nash, J., & Argyropoulos, S. (1999). Mechanisms of action of selective serotonin reuptake inhibitors in the treatment of psychiatric disorders. *European neuropsychopharmacology*, *9*, S81-S86.

Páleníček, T., Hliňák, Z., Bubeníková-Valešová, V., Novák, T., & Horáček, J. (2010). Sex differences in the effects of N, N-diethyllysergamide (LSD) on behavioural activity and prepulse inhibition. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, *34*(4), 588-596.

Pereira, J. K. D., Vieira, R. J., Konishi, C. T., De, R., & Frussa-Filho, R. (1999). The phenomenon of "one-trial tolerance" to the anxiolytic effect of chlordiazepoxide in the elevated plus-maze is abolished by the introduction of a motivational conflict situation. *Life Sciences, 65(*10), PL101-PL107. https://doi.org/10.1016/s0024-3205(99)00339-2

Pigott, H. E. (2015). The STAR* D trial: it is time to reexamine the clinical beliefs that guide the treatment of major depression. *The Canadian Journal of Psychiatry*, *60*(1), 9-13.

Porsolt, R. D., Bertin, A., & Jalfre, M. J. A. I. P. (1977). Behavioral despair in mice: a primary screening test for antidepressants. *Archives internationales de pharmacodynamie et de therapie*, *229*(2), 327-336.

Ritchie, H. & Roser, M. (2018). Mental health. Published online at OurWorldInData.org. https://ourworldindata.org/mental-health on 22/06/2021.

Rodgers, R. J., & Johnson, N. J. T. (1995). Factor analysis of spatiotemporal and ethological measures in the murine elevated plus-maze test of anxiety. *Pharmacology, Biochemistry and Behavior,* 52(2), 297–303. https://doi.org/10.1016/0091-3057(95)00138-M

Roseman, L., Leech, R., Feilding, A., Nutt, D. J., & Carhart-Harris, R. L. (2014). The effects of psilocybin and MDMA on between-network resting state functional connectivity in healthy volunteers. *Frontiers in human neuroscience, 8,* 204.

Roseman, L., Nutt, D. J., & Carhart-Harris, R. L. (2018). Quality of acute psychedelic experience predicts therapeutic efficacy of psilocybin for treatment-resistant depression. *Frontiers in pharmacology, 8,* 974.

Ross, S., Bossis, A., Guss, J., Agin-Liebes, G., Malone, T., Cohen, B., ... & Schmidt, B. L. (2016). Rapid and sustained symptom reduction following psilocybin treatment for anxiety and depression in patients with life-threatening cancer: a randomized controlled trial. *Journal of psychopharmacology, 30*(12), 1165-1180.

Rush, A. J., Warden, D., Wisniewski, S. R., Fava, M., Trivedi, M. H., Gaynes, B. N., & Nierenberg, A. A. (2009). STAR* D. *CNS drugs*, *23*(8), 627-647.

Samorini, G. (2001). New data from the ethnomycology of psychoactive mushrooms. *International Journal of Medicinal Mushrooms*, *3*(2-3).

Schmid, Y., Gasser, P., Oehen, P., & Liechti, M. E. (2021). Acute subjective effects in LSD-and MDMA-assisted psychotherapy. *Journal of Psychopharmacology*, *35*(4), 362-374.

Schneider, L. S., Small, G. W., Hamilton, S. H., Bystritsky, A., Nemeroff, C. B., & Meyers, B. S. (1997). Estrogen replacement and response to fluoxetine in a multicenter geriatric depression trial. *The American Journal of Geriatric Psychiatry, 5(*2), 97-106.

Shepard, G. (2005). Psychoactive botanicals in ritual, religion and shamanism. *Ethnopharmacology, Encyclopedia of Life Support Systems. Oxford: UNESCO/EOLSS*.

Shinfuku, M., Kishimoto, T., Uchida, H., Suzuki, T., Mimura, M., & Kikuchi, T. (2019). Effectiveness and safety of long-term benzodiazepine use in anxiety disorders: a systematic review and meta-analysis. *International clinical psychopharmacology*, *34*(5), 211-221.

Simon, G. E., Savarino, J., Operskalski, B., & Wang, P. S. (2006). Suicide risk during antidepressant treatment. *American Journal of Psychiatry*, *163*(1), 41-47.

Spreng, R. N., & Grady, C. L. (2010). Patterns of brain activity supporting autobiographical memory, prospection, and theory of mind, and their relationship to the default mode network. *Journal of cognitive neuroscience*, *22*(6), 1112-1123.

Sramek, J. J., Murphy, M. F., & Cutler, N. R. (2016). Sex differences in the psychopharmacological treatment of depression. *Dialogues in clinical neuroscience*, *18*(4), 447.

Studerus, E., Gamma, A., Kometer, M., & Vollenweider, F. X. (2012). Prediction of psilocybin response in healthy volunteers. *PloS one*, *7*(2), e30800.

Studerus, E., Kometer, M., Hasler, F., & Vollenweider, F. X. (2011). Acute, subacute and long-term subjective effects of psilocybin in healthy humans: a pooled analysis of experimental studies. *Journal of psychopharmacology*, *25*(11), 1434-1452.

Taylor, S. E., Klein, L. C., Lewis, B. P., Gruenewald, T. L., Gurung, R. A., & Updegraff, J. A. (2000). Biobehavioral responses to stress in females: tend-and-befriend, not fight-or-flight. *Psychological review*, *107*(3), 411.

Tupper, K. W., Wood, E., Yensen, R., & Johnson, M. W. (2015). Psychedelic medicine: a re-emerging therapeutic paradigm. *Cmaj*, *187*(14), 1054-1059.

Trivedi, M. H., Rush, A. J., Wisniewski, S. R., Nierenberg, A. A., Warden, D., Ritz, L., ... & Star* D Study Team. (2006). Evaluation of outcomes with citalopram for depression using measurement-based care in STAR* D: implications for clinical practice. *American journal of Psychiatry*, *163*(1), 28-40.

Tylš, F., Páleníček, T., Kadeřábek, L., Lipski, M., Kubešová, A., & Horáček, J. (2016). Sex differences and serotonergic mechanisms in the behavioural effects of psilocin. *Behavioural Pharmacology*, *27*(4), 309–320. https://doi.org/10.1097/FBP.0000000000000198

Vollenweider, F. X., Vollenweider-Scherpenhuyzen, M. F., Bäbler, A., Vogel, H., & Hell, D. (1998). Psilocybin induces schizophrenia-like psychosis in humans via a serotonin-2 agonist action. *Neuroreport*, *9*(17), 3897-3902.

Vicens, C., Socias, I., Mateu, C., Leiva, A., Bejarano, F., Sempere, E., ... & Fiol, F. (2011). Comparative efficacy of two primary care interventions to assist withdrawal from long term benzodiazepine use: a protocol for a clustered, randomized clinical trial. *BMC family practice*, *12*(1), 1-7.

Walton, C. (2003). When demand exceeds supply for psychotherapy: evaluation of the client priority rating scale (CPRS) as a triage protocol.

Wasson, R. G. (1961). The hallucinogenic fungi of Mexico: an inquiry into the origins of the religious idea among primitive peoples. *Botanical Museum Leaflets, Harvard University*, *19*(7), 137-162.

Weissman, M. M., Bland, R. C., Canino, G. J., & Greenwald, S. (1994). The cross national epidemiology of obsessive compulsive disorder: The Cross National Collaborative Group. *The Journal of clinical psychiatry*.

Winkelman, M., & Winkelman, M. (2019). Archeological investigations of ancient psychoactive substances. *Journal of Psychedelic Studies*, *3*(2), 261-263.

Zhou, Y., Cao, Z., Yang, M., Xi, X., Guo, Y., Fang, M., ... & Du, Y. (2017). Comorbid generalized anxiety disorder and its association with quality of life in patients with major depressive disorder. *Scientific reports*, *7*(1), 1-8.

Zobel, I., Kech, S., Van Calker, D., Dykierek, P., Berger, M., Schneibel, R., & Schramm, E. (2011). Long-term effect of combined interpersonal psychotherapy and pharmacotherapy in a randomized trial of depressed patients. *Acta Psychiatrica Scandinavica*, *123*(4), 276-282.